MIL-A-82450B(OS)
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MILITARY SPECIFICATION

AUTOMATIC PILOT TEST SET

(For Torpedo Mk 46)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers requirements for procurement of one type of Automatic Pilot Test Set (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Ordnance Systems Command

DL 2540360	Automatic Pilot Test Set, and all documents listed hereon

PUBLICATIONS

Naval Ordnance Systems Command

WD 28935	Special Manufacturing Procedure for Calibration of Automatic Pilot Test Set, DL 2540360

FSC 1355

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(Copies of drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Basic requirements

3.1.1 Contractor-furnished data. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.1.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in manufacture of the automatic pilot test set shall be in accordance with requirements specified herein and with the applicable documents listed in DL 2540360.

3.2 Performance and product characteristics

3.2.1 Input power. The input power requirements of the test set shall be 115 ±15 volts alternating current (ac) at a maximum current of 7.5 amperes.

3.2.2 DC voltage outputs. The test set shall furnish ±40.0, ±15.0, and -30.0 direct current (dc) voltages to test set output terminals into specified loads. The voltage shall be adjustable when loaded, over a ±5 percent range, and be adjustable, under load, to within ±0.1 vdc of the specified nominal. With the exception of the -40 v SS line, ammeters shall be provided to monitor load current.

3.2.3 Performance requirements. The test set shall provide the following functions:

3.2.3.1 Exploder power supply. An output to drive the +90 vdc IN lines that appears as a source of +140 to +150 vdc open circuit voltage and 5 to 8 milliampere (ma) short circuit current.

3.2.3.2 4.8 kHz output voltage. A 4.8 kiloHertz (kHz) voltage adjustable from 4700 to 4900 Hertz (Hz), and with a magnitude that shall be adjustable from 24 to 28 volts root mean square (rms) when driving a load of 260 to 430 ma at 0.8 to 0.98 leading power factor. Distortion shall be 0.5 percent or less.

3.2.3.3 Pendulum signal. A simulated pendulum signal, continuously variable from 25.0 millivolts (mv) rms to 11.60 ±0.05 volts rms, in or

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out of phase. The signal shall have a maximum distortion of 0.5 percent greater than the 4.8 kHz output voltage. The phase relationship, referenced to the 4.8 kHz output voltage (3.2.3.2), shall be as shown in Table I.

3.2.3.4 Rate gyro signal. A simulated rate gyro signal, continuously variable from 25.0 mv rms to 11.35 ±0.05 volts rms, in or out of phase. The signal shall have a maximum distortion of 0.5 percent greater than the 4.8 kHz output voltage. The phase relationship, referenced to the 4.8 kHz output voltage (3.2.3.2), shall be as shown in Table I.

Table I

Phase Relationships

Gyro Pendulum Assembly Simulator Selector Switch	Control	Position	Phasing
PITCH	PENDULUM	DOWN	0 ±2°
	PENDULUM	UP	180 ±2°
	RATE GYRO	DOWN	0 ±2°
	RATE GYRO	UP	180 ±2°
ROLL	PENDULUM	CCW	180 ±2°
	PENDULUM	CW	0 ±2°
	RATE GYRO	CCW	180 ±2°
	RATE GYRO	CW	0 ±2°
YAW RATE-ROLL PENDULUM	PENDULUM	CW	0 ±2°
	PENDULUM	CCW	180 ±2°
	RATE GYRO	RIGHT	0 ±2°
	RATE GYRO	LEFT	180 ±2°

3.2.3.5 Simulated signals. Simulated signals shall be provided to the appropriate autopilot input line as shown in Table II. Functions and switch positions preceded by an asterisk (*) pertain to automatic pilots used with Torpedo Mk 46 Mod 2.

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Table II

Simulated Signals

Signal	Condition	Volts dc or ohms
PITCH ENABLE	OFF	0.0 ±0.1 vdc
	ON	-12.25 ±0.25 vdc
3 MINUTE SNAKE	OPEN	500,000 ohms min
	GRD	0.5 ohms max
C G ACTIVATE	OPEN	500,000 ohms min
	GRD	0.5 ohms max
RECOVERY	ON	0.5 ohms max
	OFF	500,000 ohms min
END SNAKE * INHIBIT SEARCH	OFF	0.0 ±0.1 vdc
	ON	-12.25 ±0.25 vdc
REINSERT PENDULUM * PURSUIT HOLD	OFF * ON	-12.25 ±0.25 vdc
	ON * OFF	0.0 ±0.1 vdc
PITCH INHIBIT (MOD 1 - MOD 0)	ON (MOD 1)	-21.95 ±0.55 vdc
	OFF (MOD 1)	-1.35 ±0.45 vdc
	OFF OR ON (MOD 0)	500,000 ohms min
EXPLODER ARM	OFF	±40.0 ±0.3 vdc
	ON	-40.0 ±0.3 vdc

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Table II (cont'd)

Simulated Signals

Signal	Condition	Volts dc or ohms
R<B	OPEN	500,000 ohms min
* CEILING CHANGE	GRD	0.5 ohms max
* NOT SNAKE	* ON	-15.0, +0.2, -0.3 vdc
	* OFF	0.0 ± 0.1 vdc
* NOT 6°	* ON	-15.0, +0.2, -0.3 vdc
	* OFF	0.0 ± 0.1 vdc
* NOT 10°	* ON	500,000 ohms min
	* OFF	0.5 ohm max

3.2.3.6 Course gyro. A simulated course gyro resistance and course gyro voltage shall be provided as follows:

(a) Resistance shall be variable from less than 6 ohms to 7500 ±75 ohms, with a 2.5% overtravel on each end of adjustment for null purposes.

(b) Voltage shall be continuously variable from a maximum null of 25 millivolts rms to 26.0 ±0.1 volts rms.

3.2.3.7 Depth pressure transducer. A simulated depth transducer signal shall be provided which is derived from a potentiometer of greater than 5000 ohms resistance and is continuously variable from a maximum null voltage of 25 millivolts to 26.0 ±0.1 volts rms.

3.2.3.8 20', 50', and 100' pressure switches. Simulated 20', 50', and 100' pressure switch output signals shall be provided. A simulated closed switch condition shall be -29.5 ±0.5 vdc. The simulated open condition shall be an open circuit. The simulated switches shall be combined with the COURSE GYRO or DEPTH TRANSDUCER simulator output pot in a COURSE/DEPTH SIMULATOR switch to give the outputs of Tables III, IV, and V for the indicated switch positions.

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3.2.3.9 Fire control signals. A simulated fire control voltage shall be provided that is adjustable from 18 to 30 vdc when driving a 12 ohm load. Simulated search depth and ceiling mode signals shall be provided as shown in Tables VI and VII for 20.0 ±0.1 vdc fire control voltage. Indicator lamps shall be provided to indicate when the Autopilot is preset properly and to indicate system continuity. A switch to supply the fire control voltage to the CG UNCAGE input line and test points to monitor the PISTON MOTOR signals shall be provided.

Table III

20' Switch Indications

COURSE/DEPTH SIMULATOR Selector Switch Position	Simulated 20' Switch Output
COURSE GYRO	-29.5 ±0.5
DEPTH PRESSURE XDUCER	Open
0'	Open
D>20'	-29.5 ±0.5
D>50'	-29.5 ±0.5
D>100'	-29.5 ±0.5

Table IV

50' Switch Indications

COURSE/DEPTH SIMULATOR Selector Switch Position	Simulated 50' Switch Output
COURSE GYRO	-29.5 ±0.5
DEPTH PRESSURE XDUCER	Open
0'	Open
D> 20'	Open
D> 50'	-29.5 ±0.5
D> 100'	-29.5 ±0.5

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Table V

100' Switch Indications

COURSE DEPTH SIMULATOR Selector Switch Position	Simulated 100' Switch Output
COURSE GYRO	-29.5 ±0.5
DEPTH PRESSURE XDUCER	Open
0'	Open
D > 20'	Open
D > 50'	Open
D > 100'	-29.5 ±0.5

Table VI

Search Depth Voltages

SEARCH DEPTH	Volts dc
1	+2.85 ±0.05
2	+5.7 ±0.1
3	+8.55 ±0.15
4	+11.4 ±0.2
5	+14.25 ±0.25
6	+17.1 ±0.2
7	+20.0 ±0.2

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Table VII

Ceiling Mode Voltages

CEILING MODE	Volts dc
1	+4.0 ±0.1
2	$+8.00\ ^{+0.10}_{-0.15}$
3	+12.0 ±0.2
4	+16.0 ±0.2
5	+20.0 ±0.2

3.2.3.10 The following indicators are provided which shall illuminate when -15 vdc is applied to their respective inputs:

(a) SNAKE SEARCH

(b) POWER REDUCTION

(c) DOPPLER LOGIC

(d) SD NOT ACHIEVED-PASSIVE

(e) ACTIVE

(f) PITCH RANGE CONSTR

(g) ODN TRIGGER

(h) BEAM WIDTH CONTROL

(i) COMPUTER ENABLE

(j) NOT AIR/ASROC SEARCH

(k) AIR/ASROC SEARCH

3.2.3.10.1 MOD 2 indicator shall illuminate when -25.0 vdc is applied through a 4100 ohm resistor to connector pin J4201-D simultaneously with AUTOPILOT POWER being applied.

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3.2.3.11 Autopilot output indicators shall be provided which shall function as follows:

(a) The CW, DOWN, and RIGHT indicators shall illuminate when a 75,000 ohm load is connected between their inputs and the -40 vdc terminal.

(b) The CCW, UP, and LEFT indicators shall illuminate when a 75,000 ohm load is connected between their inputs and ground.

3.2.3.12 Output loads. A switch shall be provided to select YAW, PITCH, and ROLL output loads corresponding to MOD 1 or MOD 0 configuration.

(a) With the MOD switch in the 1 position, the output loads shall be 910 ohms ± 1 percent.

(b) With the MOD switch in the 0 position, the output loads shall be 350 ohms ±5 percent.

3.2.3.13 Pitch command signals. The following simulated pitch command signals shall be provided to the UP or DOWN COMMAND inputs:

(a) 0(zero): -25 ±1 mv (Mod 1), 0±5 mv (MOD 0)

(b) 1 UP: Adjustable from +1.23 to +2.22 vdc

(c) 2 UP: Adjustable from +2.56 to +3.26 vdc

(d) 1 DOWN: Adjustable from -1.23 to -2.22 vdc

(e) 2 DOWN: Adjustable from -2.56 to -3.26 vdc

3.2.3.14 Yaw command signals. The following simulated yaw command signals shall be provided to the LEFT and RIGHT COMMAND inputs.

(a) 0(zero): -25 ±1 mv (MOD 1), 0±5 mv (MOD 0)

(b) 1 LEFT: Adjustable from -1.23 to -2.22 vdc

(c) 2 LEFT: Adjustable from -2.56 to -3.26 vdc

(d) 1 RIGHT: Adjustable from +1.23 to +2.22 vdc

(e) 2 RIGHT: Adjustable from +2.56 to +3.26 vdc

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3.2.3.15 Function selector. A selector switch shall be provided which allows monitoring of the following autopilot signals:

DOPPLER LOGIC	PITCH RANGE CONSTRAINT	NOT AIR/ASROC SEARCH
SD NOT ACHIEVED - PASSIVE	BEAM WIDTH CONTROL	AIR/ASROC SEARCH
ACTIVE	COMPUTER ENABLE	-30 V LOG AMP

3.2.3.16 400 Hertz functions. A 65 ohm load shall be provided as a dummy load for the Autopilot 400 Hz supply. An indicator to sense the presence of 400 Hertz voltage and a switch to allow application or removal of +40 vdc SS power to the 400 Hertz supply shall be provided.

3.2.3.17 Instrumentation test points. Instrumentation test points shall be provided to allow monitoring of the following Autopilot instrumentation lines:

INST COURSE GYRO	INST PITCH RATE GYRO
INST ROLL RATE GYRO	INST ROLL PENDULUM
INST YAW RATE GYRO	INST PITCH PENDULUM

3.3 Workmanship. Workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. Inspections of the test set shall be classified as follows:

4.2.1 Quality conformance inspection

(a) Quality conformance sampling (see 4.3.1)

(b) Quality conformance tests and examinations (see 4.5)

4.3 Sampling

4.3.1 Quality conformance sampling. The tests and examinations specified in 4.5 shall be conducted on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the test set shall be subjected to tests under the following conditions:

(a) Temperature	Room ambient (60 to 95 degrees Fahrenheit (F))
(b) Humidity (relative)	Room ambient, up to 95 percent
(c) Altitude	Normal ground atmospheric pressure

4.5 Quality conformance tests and examinations. Unless otherwise specified, each test set delivered under contract shall be subjected to the tests and examinations contained in 4.5.3.

4.5.1 Conditions

4.5.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in the order shown.

4.5.1.2 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits are given in Limits column. The individual performing the test or inspection shall initial in space provided in the Reading column, indicating that requirements were met satisfactorily.

4.5.1.3 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the

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contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

4.5.2 Quality conformance test equipment. Items of test equipment shown in Table VIII are required to perform the quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals and procedures for the results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary.

Table VIII

Quality Conformance Test Equipment

Item	Identification
Digital Voltmeter - (Range: 0.001 to 999.9 vdc or vrms: dc accuracy: ±0.05% of reading ±1 digit; ac accuracy: ±0.1% of reading ±2 counts; Input impedance: 10 megohms, 20 pf.)	Hewlett-Packard Model 3440A with 3445A AC/DC Range Unit
Oscilloscope-(Bandpass: from DC to 15 megaHertz; Rise time: from 0.023 microsecond; Accuracy: 3 percent from 0.05 to 20 volts/cm.)	Tektronix Model 535A with type CA preamplifier
Resistance Bridge-(Range: 0.001 to 9,999,000 ohms; Accuracy: 0.15 percent ±0.01 ohm from 1.0 to 40,000 ohms)	Leeds & Northrup No. 5305 Test Set
Distortion Analyzer-(Total harmonic distortion: ±3% over frequency range of 20 Hz to 20 kHz.)	Hewlett-Packard Model 330B
Phase Meter-(Frequency Range: 15 Hz to 500 megaHertz. Full Scale Sensitivity: 1° or 0.04 volt rms. Phase Angle Range: 0-1, 0-2, 0-4, 0.05° or 2%.)	Ad-Yu Vectorlyzer, Model 202

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Table VIII

Quality Conformance Test Equipment (cont'd)

Item	Identification
Resistors (Loads)-(1 - 75,000 ohms ±5%, 1/2W 2 - 130,000 ohms ±5%, 1/2W 1 - 200 ohms ±1%, 10W 1 - 80 ohms ±1%, 50W 1 - 60 ohms ±1%, 10W 1 - 12 ohms ±5%, 150W) 1 - 4700 ohms ±2% 1/2W 1 - 35,700 ohms ±1% 1/4W	Commercial
Electronic Counter-(Range: from 10 Hz to 120 kHz; Accuracy: to ±1 count.)	Hewlett-Packard Model 522B
Multimeter - (Ohmmeter; Resistance range: 0 to 20 megohms; Accuracy: within 3° of arc on resistance scale.)	Simpson Model 260
Diode, IN645	Commercial

4.5.3 Test procedures

4.5.3.1 DC power supply

NOTE: Before turning on the 4.8 KC ON-OFF switch (PANEL NO. 3), jumper pin J4201-X to pins J4202-J and J4201-P, and pin J4201-V to pin J4202-D. These jumpers shall remain connected unless otherwise specified.

4.5.3.1.1 Preliminary setup

(a) Turn 4.8 KC ON-OFF switch off. Turn on all power supply power switches. Turn ON-OFF switch (POWER PANEL) and AUTOPILOT POWER switch to ON. Permit equipment to stabilize for 15 minutes before taking data. On digital voltmeter, set FUNCTION switch to DC and RANGE switch to AUTO.

NOTE: While connecting or disconnecting load resistors specified throughout following tests, AIR/ASROC-TUBE LAUNCH switch shall be set at AIR/ASROC. Set same switch to OFF while making voltage and current measurements.

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NOTE: All voltages of Table IX shall be measured at appropriate test jacks located on PANEL NO. 1, using digital voltmeter. Current readings shall be made after power supply voltages have been adjusted to specified voltage indications with specified loads connected. Adjustment controls for +15 vdc and -30 vdc supplies are located on test PANEL NO. 1.

NOTE: After each voltage reading in Table IX, set AIR/ASROC - TUBE LAUNCH switch to TUBE LAUNCH and then AIR/ASROC. Voltage indication in both positions shall be -0.2 to +0.2 volt dc.

(b) Connect resistances listed in LOAD RES. column of Table IX between J4201-P (GRD) and connector pins listed.

Table IX

Adjustment of Power Supplies

Power Supply (Volts DC)	Connector and Pin	Load Res. (Ohms)	Indication (Volts DC)	Current Meter Indication
+40	J4201-U	80 (50W)	+39.9 to +40.1	480 to 520 ma
-40	J3601-JJ	Infinite	-39.9 to -40.1	
-15	J3601-LL	60 (10W)	-14.9 to -15.1	238 to 262 ma
+15	J3601-KK	200 (10W)	+14.9 to +15.1	71 to 77 ma
-30	J3601-NN	200 (10W)	-29.9 to -30.1	142 to 157 ma

4.5.3.1.2 +40 VDC SS supply

(a) Set 400~ switch on PANEL NO. 1 to ON. Adjust +40 VDC supply to +40.0 ±0.1 (+39.9 to +40.1) volts dc as indicated by digital voltmeter.

(b) Current meter indication shall be as shown in Table IX.

(c) Set 400~ switch to OFF and remove load resistor.

4.5.3.1.3 -40 VDC SS supply

(a) Adjust -40 VDC power supply to -40.0 ±0.1 (-39.9 to -40.1) volts dc, as indicated by digital voltmeter.

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4.5.3.1.4 -15 VDC supply

(a) Adjust -15 VDC power supply to -15.0 ±0.1 (-14.9 to -15.1) volts dc, as indicated by digital voltmeter.

(b) Current indication shall be as shown in Table IX.

(c) Remove load resistor.

4.5.3.1.5 +15 VDC supply

(a) Check supply for adjustment range of +14.25 to +15.75 volts dc.

(b) Adjust +15 VDC power supply to +15.0 ±0.1 (+14.9 to +15.1) volts dc, as indicated by digital voltmeter.

(c) Current indication shall be as shown in Table IX.

(d) Remove load resistor.

4.5.3.1.6 -30 VDC supply

(a) Check supply for adjustment range of -28.5 to -31.5 volts dc.

(b) Adjust -30 VDC power supply to -30.0 ±0.1 (-29.9 to -30.1) volts dc, as indicated by digital voltmeter.

(c) Current indication shall be as shown in Table IX.

(d) Remove load resistor.

4.5.3.1.7 +125 VDC supply

(a) Connect two 130K, 1/2 watt resistors (±5%) from J4201-S (+90 VDC IN) and J4201-T (+90 VDC IN) to J4201-P (GRD). Connect digital voltmeter to the +125 VDC test point on PANEL NO. 1.

(b) Adjust +125 VDC ADJ control to obtain +125 ±2 (+123 to +127) volts dc indication on digital voltmeter.

(c) Connect digital voltmeter to +90 VDC IN (S) and +90 VDC IN (T) test points.

(d) Digital voltmeter shall indicate +110 ±3 (+107 to +113) volts dc at each test point.

(e) Remove the 130K load resistors.

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4.5.3.2 Internal test set voltages

(a) Connect digital voltmeter between +5 VDC TS test point and PWR GND test point located inside test set cabinet on Chassis No. 3. Indication shall be +5.0 ±0.4 (+4.6 to +5.4) volts dc.

(b) Connect digital voltmeter between -20 VDC TS test point and PWR GND test point. Indication shall be -20.0 ±1.3 (18.7 to -21.3) volts dc.

4.5.3.3 Test point wiring

(a) Using digital voltmeter, measure voltage at +40 VDC TS, +15 VDC TS, -15 VDC TS, -30 VDC TS, -40 VDC TS test points on Chassis No. 3. Indications shall be approximately the same as those recorded for 4.5.3.1.2 through 4.5.3.1.6. Variations in indications will be because supplies are no longer loaded.

(b) Using digital voltmeter, measure voltage at +150 VDC TS test point on Chassis No. 3. Indication shall be +140 to +160 vdc. Set FUNCTION switch on digital voltmeter to AC.

4.5.3.4 Indicator lamp pretest

(a) With AUTOPILOT POWER switch ON, check press-to-test capability of all indicator lamps.

(b) All indicators shall illuminate when pressed.

4.5.3.5. 4.8 KC power supply

4.5.3.5.1 Amplitude adjustment

(a) Connect digital voltmeter to 4.8 KC test points on PANEL NO. 3. Jumper pins J4201-X (4.8 KC GRD), J4202-J (common) and J4201-P (GRD); and also J4201-V (4.8 KC IN ∅) and J4202-D (4.8 KC IN ∅). Turn on 4.8 KC ON-OFF switch. Allow 5 minutes warmup time.

(b) Set 4.8 KC VOLTAGE ADJ on PANEL NO. 3 to mid-range.

(c) Set AUTOPILOT POWER switch to ON.

(d) Set FREQ SELECT switch (on PANEL NO. 4 chassis, inside cabinet) to 4.8 KC.

(e) Adjust 4.8 KC VOLTAGE ADJ control to verify an adjustment range to 24.0 to 28.0 volts rms on digital voltmeter. Record maximum and minimum indications.

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4.5.3.5.2 4.9 KC adjustment

(a) Set 4.8 KC amplitude to 26.0 ±0.1 volts rms. Connect electronic counter to J4201 pins V (4.8 KC In ∅) and X (4.8 KC GRD).

(b) Set OSC FREQUENCY SELECTOR to 4.9 KC. Frequency shall be adjustable, using 4.8 KC FREQUENCY ADJ on PANEL NO. 3, from 4890 to 4910 Hz. Record maximum and minimum indications.

4.5.3.5.3 4.8 KC adjustment

(a) Set OSC FREQUENCY SELECTOR to 4.8 KC. Frequency shall be adjustable from 4790 Hz to 4810 Hz. Record maximum and minimum indications.

4.5.3.5.4 4.7 KC adjustment

(a) Set OSC FREQUENCY SELECTOR to 4.7 KC. Frequency shall be adjustable from 4690 to 4710 Hz. Record maximum and minimum indications.

4.5.3.5.5 4.8 KC readjustment

(a) Set OSC FREQUENCY SELECTOR to 4.8 KC and adjust frequency, using FREQUENCY ADJ control, to 4800 Hz ±2 Hz. Remove electronic counter from test circuit.

4.5.3.6 Gyro pendulum assembly simulator

(a) Connect digital voltmeter leads to connector pins J4202-T (pitch pendulum) and J4201-P (GRD).

(b) Set GYRO PENDULUM ASSEMBLY SIMULATOR SELECTOR to PITCH.

(c) Turn PENDULUM Helidial toward 0.00 until a null appears on digital voltmeter. Set Helidial to 0.00 at point of minimum voltage indication on digital voltmeter. With Helidial at 0.00, maximum indication shall be 25.0 millivolts.

(d) Turn PENDULUM Helidial CW to 5.00.

(e) Digital voltmeter shall indicate 11.55 to 11.65 volts rms (11.60 volts rms ±50 millivolts).

(f) Rotate PENDULUM Helidial CCW to 5.00. Digital voltmeter shall indicate 11.55 to 11.65 volts rms (11.60 volts rms ±50 millivolts).

(g) Connect digital voltmeter leads to connector pins J4202-S (pitch rate gyro) and J4201-P (GRD).

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(h) Turn RATE GYRO Helidial toward 0.00 until a null appears on digital voltmeter. Set Helidial to 0.00 at null. With Helidial at 0.00, maximum indication shall be 25.0 millivolts.

(i) Rotate RATE GYRO Helidial CW to 5.00. Digital voltmeter shall indicate 11.30 to 11.40 volts rms (11.35 volts rms ±50 millivolts).

(j) Rotate RATE GYRO Helidial CCW to 5.00. Digital voltmeter shall indicate 11.30 to 11.40 volts rms (11.35 volts ±50 millivolts).

(k) Connect distortion analyzer to 4.8 KC test points on PANEL NO. 3. Distortion indication shall be 0.5% maximum.

(l) With PENDULUM Helidial turned CW to 5.00, connect distortion analyzer to pins J4202-T (Pitch Pendulum) and J4201-P (GRD). Record indicated distortion. Distortion content shall not be more than 0.5% greater than distortion measured at 4.8 KC test points on PANEL NO. 3. Record difference in distortion indications. Remove distortion analyzer.

(m) Connect phase meter first input to connector pins J4202-D (4.8 KC IN Ø) and J4201-X (GRD), second input to J4201-X (GRD) and J4202 pins listed in Table X. Set Helidials to 5.00 in the direction indicated. Determine phase relationship according to condition of Table X as indicated by phase meter. Record phase relation indication.

(n) Connect oscilloscope channel B to 4.8 KC IN Ø test point on chassis No. 3 inside test set, channel A to IN Ø SIM test point using SIM GRD for oscilloscope common. Test point voltage waveforms shall be in phase.

(o) Connect oscilloscope channel A to OUT Ø SIM test point. Channel A and B test point voltage waveforms shall be 180° out of phase.

(p) Set AUTOPILOT POWER switch to OFF. Both channel A and B indications shall change to an approximate zero volts.

(q) Remove test equipment and jumpers from connector pins J4202-D (4.8 KC IN Ø) and J4202-J (common).

Table X

Phase Relationships

Test	Gyro Pendulum Assembly Simulator Selector	Helidial	Helidial Position	Connector J4202 Pin	Phasing & Limits
1	PITCH	PENDULUM	CW	T	0° ±2°
2		PENDULUM	CCW	T	180° ±2°

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Table X con't.

Phase Relationships

Test	Gyro Pendulum Assembly Simulator Selector	Helidial	Helidial Position	Connector J4202 Pin	Phasing & Limits
3		RATE GYRO	CW	S	0° ±2°
4		RATE GYRO	CCW	S	180° ±2°
5	ROLL	PENDULUM	CCW	L	180° ±2°
6		PENDULUM	CW	L	0° ±2°
7		RATE GYRO	CCW	U	180° ±2°
8		RATE GYRO	CW	U	0° ±2°
9	YAW RATE-ROLL PENDULUM	PENDULUM	CW	L	0° ±2°
10		PENDULUM	CCW	L	180° ±2°
11		RATE GYRO	CW	K	0° ±2°
12		RATE GYRO	CCW	K	180° ±2°

4.5.3.7 Course depth simulator

(a) Connect jumpers between J4201-V, J4202-D, and P3602-A (4.8 KC IN ∅), and between P3602-B, J4202-J, J4201-P, and J4201-X (GRD).

(b) Connect digital voltmeter leads to COURSE/DEPTH test points located beneath COURSE/DEPTH Helidial.

(c) Set AUTOPILOT POWER switch to ON.

(d) Set COURSE/DEPTH-SIMULATOR SELECTOR switch to COURSE GYRO.

(e) Rotate COURSE/DEPTH Helipot slowly CCW while observing digital voltmeter. Stop Helipot adjustment at point digital voltmeter indicates null. Set Helidial to 0.00. Digital voltmeter shall indicate 25 millivolts maximum.

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(f) Rotate COURSE/DEPTH Helidial to 10.00. Digital voltmeter shall indicate 26 ±0.1 volts rms.

(g) Set AUTOPILOT POWER and 4.8 KC ON-OFF switches to OFF and remove jumpers. Remove test equipment.

(h) Connect resistance bridge to connector pins P3602-C (Course Gyro) and P3602-B (4.8 KC GRD).

(i) With COURSE/DEPTH Helipot at 0.00, resistance bridge meter shall indicate less than 6 ohms.

(j) Rotate COURSE/DEPTH Helidial to 10.00. Resistance bridge shall indicate 7425 to 7575 ohms (7500 ±75 ohms).

(k) Set COURSE/DEPTH SIMULATOR SELECTOR to DEPTH-PRESSURE XDUCER. Connect resistance bridge between connector pins J4202-P (Simulator Xducer) and J4201-P (GRD). Resistance bridge shall indicate 95 to 105 ohms.

(l) Remove resistance bridge. Turn AUTOPILOT POWER switch to ON. Set COURSE/DEPTH SIMULATOR SELECTOR to COURSE GYRO. Set FUNCTION switch on digital voltmeter to DC. Adjust the -30 V power supply to -30.0 ±0.1 (29.9 to -30.1) volts dc, as indicated by digital voltmeter.

(m) Using digital voltmeter, measure voltage at D≥ 20' test points while the SIMULATOR SELECTOR is rotated to the indicated positions. Voltage indications shall be as specified in Table XI.

Table XI

D≥ 20' Voltage Indications

Test	COURSE/DEPTH SIMULATOR SELECTOR Positions	D≥20' Volts DC Limits
1	COURSE GYRO	-29.0 to -30.0
2	DEPTH PRESSURE XDUCER	-1.0 to +1.0
3	0'	-1.0 to +1.0
4	D≥20'	-29.0 to -30.0
5	D≥50'	-29.0 to -30.0
6	D≥100'	-29.0 to -30.0

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(n) Set COURSE/DEPTH SIMULATOR SELECTOR TO D≥50'. Using digital voltmeter, measure voltage at D≥50' test points, while the SIMULATOR SELECTOR is rotated to the indicated positions. Voltage indications shall be as specified in Table XII.

Table XII

D≥50' Voltage Indications

Test	COURSE/DEPTH SIMULATOR SELECTOR Positions	D≥50' Volts DC Limits
1	COURSE GYRO	-29.0 to -30.0
2	DEPTH PRESSURE XDUCER	-1.0 to +1.0
3	0'	-1.0 to +1.0
4	D≥20'	-1.0 to +1.0
5	D≥50'	-29.0 to -30.0
6	D≥100'	-29.0 to -30.0

(o) Set COURSE/DEPTH SIMULATOR SELECTOR to D≥100'. Using digital voltmeter, measure voltage at connector pins J4202-F (D≥ 100') and J4201-P (GRD) while the SIMULATOR SELECTOR is rotated to the indicated positions. Voltage indications shall be as specified in Table XIII.

Table XIII

D≥100' Voltage Indications

Test	COURSE/DEPTH SIMULATOR SELECTOR Positions	D≥ 100' Volts DC Limits
1	COURSE GYRO	-29.0 to -30.0
2	DEPTH PRESSURE XDUCER	-1.0 to +1.0
3	0'	-1.0 to +1.0
4	D≥20'	-1.0 to +1.0
5	D≥50'	-1.0 to +1.0
6	D≥100'	-29.0 to -30.0

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(p) Remove digital voltmeter from test circuit and set COURSE DEPTH SIMULATOR SELECTOR to COURSE GYRO. Set AUTOPILOT POWER switch to OFF.

4.5.3.8 Fire control

CAUTION: The FIRE CONTROL VOLTAGE ADJ COM and FIRE CONT COM (grey test points) have a potential of -20 vdc. Do not use them for general commons or jumper them to other commons. Use only for fire control test and measurements.

CAUTION: Verify that digital voltmeter common is floating and not tied to chassis ground.

(a) Connect 12 ohm ±5% 150 watt load resistor between J4201 pins m (TUBE LAUNCH) and g (FIRE CONTROL COM). Set AIR/ASROC-Tube LAUNCH switch located on PANEL NO. 2 to TUBE LAUNCH.

(b) Using digital voltmeter, measure fire control voltage at FIRE CONTROL VOLTAGE ADJ test points on front panel. Fire control voltage shall be adjustable (using FIRE CONTROL VOLTAGE ADJ) from 18 to 30 volts dc. Record minimum and maximum indications. Set AIR/ASROC-TUBE LAUNCH switch to OFF. Remove load resistor.

(c) Set AIR/ASROC-TUBE LAUNCH switch to TUBE LAUNCH. Set FIRE CONTROL VOLTAGE ADJ for an indication of +20.0 ±0.1 volts dc.

(d) Connect digital voltmeter between connector J4201 pins i (SEARCH DEPTH SENSOR) and g (FIRE CONTROL COMMON).

(e) Set DRIVE-TEST switch to DRIVE; SEARCH DEPTH switch to position 1.

NOTE: Meter indications may vary due to ac line voltage variations. The following procedure will ascertain linear incremental voltage change for each SEARCH DEPTH switch position. If measurements fail to meet specifications, ascertain that fire control voltage is +19.9 to +20.1 (+20.0 ±0.1) volts dc.

(f) Record digital voltmeter indications for Table XIV positions of the SEARCH DEPTH switch. Indications shall be within limits listed.

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Table XIV

Search Depth Voltages

Test	SEARCH DEPTH Switch Positions	Limits (Volts DC)
1	1	+2.8 to +2.9
2	2	+5.6 to +5.8
3	3	+8.4 to +8.7
4	4	+11.2 to +11.6
5	5	+14.0 to +14.5
6	6	+16.9 to +17.3
7	7	+19.8 to +20.2

(g) Depress BRIDGE switch on FIRE CONTROL panel. Digital voltmeter shall indicate -1.0 to +1.0 vdc.

(h) Set AIR/ASROC-TUBE LAUNCH switch to AIR/ASROC. Voltage indications for SEARCH DEPTH switch position 4 shall be +11.2 to +11.6 vdc.

(i) Connect digital voltmeter between SEARCH DEPTH and FIRE CONT COM test points on PANEL NO. 2. Voltage indication for SEARCH DEPTH switch position 7 shall be +19.8 to +20.2 vdc. Set AIR/ASROC-TUBE LAUNCH switch to OFF.

(j) Connect 200-ohm, 10 watt load resistance between connector pins J4201-i (SEARCH DEPTH SENSOR) and J4201-g (FIRE CONTROL COMMON). Connect digital voltmeter to connector pins J4201-j (SEARCH DEPTH DRIVE) and J4201-g (FIRE CONTROL COMMON).

(k) Set AIR/ASROC-TUBE LAUNCH switch to TUBE LAUNCH. Depress SET SEARCH DEPTH switch. Voltage indication shall be +19.8 to +20.2 (20.0 ±0.2) volts dc and SEARCH DEPTH indicator lamp shall illuminate. Voltmeter shall indicate for a maximum of seven seconds before dropping to zero due to release by time delay relay K206.

(l) Return AIR/ASROC-TUBE LAUNCH switch to OFF. Remove 200-ohm load resistor. Connect digital voltmeter between connector pins J4201-n (CEILING SENSOR) and J4201-g (FIRE CONTROL COMMON).

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(m) Set AIR/ASROC-TUBE LAUNCH switch to TUBE LAUNCH.

(n) Digital voltmeter indications for Table XV positions of CEILING MODE switch shall be within limits listed.

Table XV

Ceiling Mode Voltages

Test	CEILING MODE Switch Positions	Limits (Volts DC)
1	1	+3.9 to +4.1
2	2	+7.85 to +8.10
3	3	+11.8 to +12.2
4	4	+15.8 to +16.2
5	5	+19.8 to +20.2

(o) Connect digital voltmeter between CEILING MODE test point and FIRE CONT COM test point on PANEL NO. 2. With CEILING MODE switch in position 5 the voltmeter shall indicate +19.8 to +20.2 vdc.

(p) Depress BRIDGE switch. Digital voltmeter shall indicate -1.0 to +1.0 volt dc.

(q) Set AIR/ASROC-TUBE LAUNCH switch to OFF. Connect 200-ohm, 10 watt load resistance between connector J4201 pins n (CEILING SENSOR) and g (FIRE CONTROL COMMON). Connect digital voltmeter to connector J4201 pins k (CEILING DRIVE) and g (FIRE CONTROL COMMON).

(r) Set AIR/ASROC-TUBE LAUNCH switch to TUBE LAUNCH. Depress SET CEILING MODE switch. Digital voltmeter shall indicate +19.8 to +20.2 (+20 ±0.2) volts dc. SET CEILING MODE indicator lamp shall be illuminated. Voltmeter shall indicate for a maximum of seven seconds before dropping to zero due to release by time delay relay K206.

(s) Set AIR/ASROC-TUBE LAUNCH switch to OFF. Remove 200-ohm load resistor. Connect digital voltmeter to connector J3601 pins UU (C.G. UNCAGE) and L (FIRE CONTROL COMMON).

(t) Set AIR/ASROC-TUBE LAUNCH switch to AIR/ASROC. Depress C.G. UNCAGE button. Digital voltmeter shall indicate +19.8 to +20.2 (+20 ±0.2) volts dc.

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(u) Connect resistance bridge to PISTON MOTOR test points on PANEL NO. 2. Resistance indications shall be between 1900 and 2100 ohms. Remove resistance bridge leads.

(v) Return AIR/ASROC-TUBE LAUNCH switch to OFF. Remove digital voltmeter leads.

4.5.3.9 Functional checks

(a) Connect a 1N645 diode between J3601-LL (-15 VDC) and J4201-D (PASSIVE). See Figure 1. Set AUTOPILOT POWER switch to ON. MOD 2 indicator shall not illuminate. Remove diode and any connecting leads.

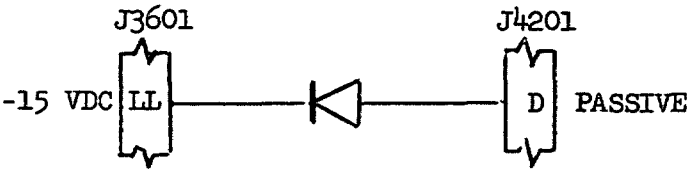


Figure 1

(b) Using a jumper wire with one end connected to connector pin J3601-LL (-15 vdc), touch free end momentarily to pins listed in Table XVI. The indicator lamps shall illuminate as specified.

(c) Touch jumper to connector J4201 pins f (COMMON and CONTINUITY) and g (FIRE CONTROL COMMON). CONTINUITY indicator located on PANEL NO. 2 shall illuminate.

4.5.3.10 Autopilot outputs

(a) Connect 75,000-ohm ½ watt load resistor between pins listed in Table XVII. Indicator lamps shall function as indicated.

Table XVI

Function Indicator Lamps

Test	Connector and Pin	Indicator Lamp Illuminates
1	J3601-C	SNAKE SEARCH
2	J4201-M	POWER REDUCTION
3	J4201-L	DOPPLER LOGIC

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Table XVI cont.

Function Indicator Lamps

Test	Connector and Pin	Indicator Lamp Illuminates
4	J4201-D	SD NOT ACHIEVED - PASSIVE
5	J4201-J	ACTIVE
6	J4201-A	PITCH RANGE CONSTR
7	J4201-K	BEAM WIDTH CONTROL
8	J3601-B	COMPUTER ENABLE
9	J4201-H	NOT AIR/ASROC SEARCH
10	J4201-E	AIR/ASROC SEARCH
11	J4201-C	ODN TRIGGER

(b) Set AUTOPILOT POWER switch to OFF. Using resistance bridge, measure resistance between -40 VDC SS test point on PANEL NO. 1 and AUTOPILOT OUTPUTS test points listed in Table XVIII at both Mod 1 and Mod 0 switch positions.

(c) Disconnect resistance bridge. Set AUTOPILOT POWER switch to ON.

4.5.3.11 Pitch and Yaw commands.

NOTE: All eight potentiometer controls referred to in the following tests are 25-turn controls.

(a) Adjust +15 vdc power supply to +14.9 to +15.1 vdc. Adjust -15 vdc power supply to -14.9 to -15.1 vdc.

(b) Connect digital voltmeter to connector pins J3601-F (UP) and J4201-P (GRD). The digital voltmeter shall indicate -24 to -26 millivolts dc with the MOD switch in the 1 position and -5 to +5 millivolts dc at the 0 position.

(c) Set PITCH COMMANDS selector switch to UP 1. Set ADJ 2 potentiometer control for UP 2 to approximate center of its travel. Turn ADJ 1 potentiometer control for UP 1 counterclockwise, then clockwise. Digital voltmeter indication shall vary from +1.23 vdc maximum to +2.22 vdc minimum.

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Table XVII

Autopilot Output Indicators

Test	Connector Pins From	Connector Pins To	Lamp	Condition
1	J3601-JJ	J3601-T	CW CCW	ILLUMINATED EXTINGUISHED
2	J4201-P	J3601-T	CW CCW	EXTINGUISHED ILLUMINATED
3	J3601-JJ	J3601-V	DOWN UP	ILLUMINATED EXTINGUISHED
4	J4201-P	J3601-V	DOWN UP	EXTINGUISHED ILLUMINATED
5	J3601-JJ	J3601-U	RIGHT LEFT	ILLUMINATED EXTINGUISHED
6	J4201-P	J3601-U	RIGHT LEFT	EXTINGUISHED ILLUMINATED

Table XVIII

Autopilot Outputs Resistance

Test Point	Limits (Ohms) MOD 1	Limits (Ohms) MOD 0
YAW (output)	900 to 920	332 to 368
PITCH (output)	900 to 920	332 to 368
ROLL (output)	900 to 920	332 to 368

(d) Readjust ADJ 1 control to the approximate center of its travel.

(e) Set PITCH COMMANDS selector switch to UP 2 and turn ADJ 2 potentiometer control for UP 2 counterclockwise, then clockwise. Digital voltmeter indication shall vary between +2.56 vdc maximum to +3.26 vdc minimum.

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(f) Readjust ADJ 2 control to approximate center of its travel.

(g) Connect digital voltmeter between pins J3601-H (DOWN COMMAND) and J4201-P (GRD). Set PITCH COMMANDS selector switch to DOWN 1. Set ADJ 2 potentiometer control for DOWN 2 to approximate center of its travel. Turn ADJ 1 potentiometer control for DOWN 1 counterclockwise, then clockwise. Digital voltmeter indication shall vary between -1.23 vdc maximum to -2.22 vdc minimum.

(h) Readjust ADJ 1 control to approximate center of its travel.

(i) Set PITCH COMMANDS selector switch to DOWN 2. Turn ADJ 2 potentiometer control for DOWN 2 counterclockwise, then clockwise. Digital voltmeter indication shall vary from -2.56 vdc maximum to -3.26 vdc minimum.

(j) Readjust ADJ 2 control to approximate center of its travel.

(k) Connect digital voltmeter to connector pins J3601-K (LEFT COMMAND) and J4201-P (GRD). The digital voltmeter shall indicate -24 to -26 millivolts dc at the Mod 1 switch position and -5 to +5 millivolts dc at the Mod 0 switch position.

(l) Set YAW COMMANDS selector switch to LEFT 1. Set ADJ 2 potentiometer control for LEFT 2 to approximate center of its travel. Turn ADJ 1 potentiometer control for LEFT 1 counterclockwise, then clockwise. Digital voltmeter indication shall vary from -1.23 vdc maximum to -2.22 vdc minimum.

(m) Readjust ADJ 1 control to approximate center of its travel.

(n) Set YAW COMMANDS selector switch to LEFT 2. Turn ADJ 2 potentiometer control for LEFT 2 counterclockwise, then clockwise. Digital voltmeter indication shall vary from -2.56 vdc maximum to -3.26 vdc minimum.

(o) Readjust ADJ 2 control to approximate center of its travel.

(p) Connect digital voltmeter between pins J3601-J (RIGHT COMMAND) and J4201-P (GRD). Set YAW COMMANDS selector switch to RIGHT 1. Set ADJ 2 potentiometer control for RIGHT 2 to approximate center of its travel. Turn ADJ 1 potentiometer control for RIGHT 1 counterclockwise, then clockwise. Digital voltmeter indication shall vary from +1.23 vdc maximum to +2.22 vdc minimum.

(q) Readjust ADJ 1 control to approximate center of its travel.

(r) Set YAW COMMANDS selector switch to RIGHT 2. Turn ADJ 2 potentiometer control for RIGHT 2 counterclockwise, then clockwise. Digital voltmeter indication shall vary from +2.56 vdc maximum to +3.26 vdc minimum.

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(s) Readjust ADJ 2 control to approximate center of its travel.

(t) Set AUTOPILOT POWER switch to OFF.

(1) Connect one lead of a 4700 ohm load resistor to J3601-NN (-30 VDC).

(2) Connect one lead of a 35,700 ohm load resistor to J3601-KK (+15 VDC).

(3) Join the other leads of these two resistors together and connect the junction to J4201-D (PASSIVE). See Figure 2.

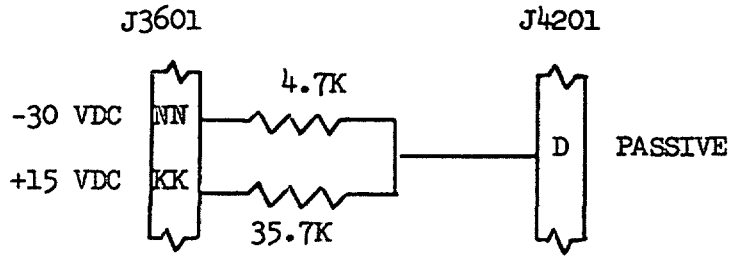


Figure 2

(u) Set AUTOPILOT POWER switch to ON. MOD 2 indicator lamp shall illuminate. Remove resistors and leads connected in 4.5.3.11 (t). MOD 2 indicator shall remain illuminated.

(v) Connect digital voltmeter negative lead to GRD (YAW) test point and positive lead to RIGHT test point. Set YAW COMMANDS switch to RIGHT 2. Set ADJ 2 (RIGHT) screw adjust control for an indication on the digital voltmeter of +2.82 ±0.02 vdc. Connect digital voltmeter positive lead to connector J3601-J (RIGHT). Digital voltmeter indication shall be +1.41 ±0.03 vdc.

(w) Connect digital voltmeter positive lead to UP test point. Set PITCH COMMANDS switch to UP 2. Set ADJ 2 (UP) screw adjust control for an indication on the digital voltmeter of +2.82 ±0.02 vdc. Connect digital voltmeter positive lead to J3601-F (UP). Digital voltmeter indication shall be +1.41 ±0.03 vdc. Disconnect digital voltmeter.

(x) Connect resistance bridge between LEFT test point and J3601-J (RIGHT). Resistance shall measure between 25,300 and 25,800 ohms. Connect resistance bridge between DOWN test point and J3601-F (UP). Resistance shall measure between 25,300 and 25,800 ohms. Disconnect resistance bridge.

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(y) Connect a jumper wire between J3601-LL (-15 VDC) and J3601-K (LEFT). PASSIVE indicator lamp shall illuminate.

(z) Set FUNCTION SELECTOR switch to SD NOT ACHIEVED - PASSIVE. Connect digital voltmeter negative lead to FUNCTION SELECTOR GRD test point and positive lead to FUNCTION SELECTOR test point. Digital voltmeter indication shall be -14.7 to -15.2 vdc. Remove jumper.

(aa) Connect digital voltmeter between J3601-F (UP) and J4201-P (GRD). Set MOD switch to 1. Voltmeter indication shall be -5 to +5 (0 ± 5) millivolts dc.

4.5.3.12 Voltage wiring check

(a) Adjust -15 vdc power supply to -14.9 to -15.1 volts; -30 vdc power supply to -29.9 to -30.1 volts.

(b) Connect digital voltmeter to connector pin J4201-P (GRD) and pins listed in succession. Set switches at bottom of PANEL 2 to the positions specified in Table XIX. At each setting, the voltage indication shall be as specified in Table XIX.

(c) Connect multimeter between J4201-P (GRD) and J4201-D (PASSIVE). Set NOT 10° switch to OFF. Resistance shall be 0.5 ohm maximum. Set NOT 10° switch to ON. Resistance shall be 500,000 ohms minimum. Disconnect multimeter.

(d) Set AUTOPILOT POWER and ON-OFF POWER switch on POWER PANEL to OFF.

4.5.3.13 Continuity tests

(a) PANEL NO. 2 switch continuity. Connect ohmmeter to connector pin J4201-P (GRD) and pins listed in Table XX. Continuity indications shall be as specified in Table XX.

(b) Function selector continuity. Connect an ohmmeter between FUNCTION SELECTOR test point and connector J4201 pins listed in Table XXI. Maximum resistance shall be as specified.

(c) Fire control test point continuity. Connect one lead of ohmmeter to connector J4201 pin g. Touch other lead to FIRE CONTROL VOLTAGE ADJ COM then to FIRE CONT COM test points on PANEL NO. 2. Resistance indication shall be between 900 and 1100 ohms at either test point.

(d) PANEL NO. 1 test point continuity. Connect ohmmeter between PANEL NO. 1 test points and connector pins listed in Table XXII. Maximum resistance shall be as specified.

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Table XIX

Voltage Wiring Check

Test	PANEL NO. 2 Switch	Switch Position	Connector J3601 Pin	Limits Volts DC
1	PITCH ENABLE	OFF	D	-0.1 to +0.1
2	PITCH ENABLE	ON	D	-12.0 to -12.5
3	END SNAKE	OFF	N	-0.1 to +0.1
4	END SNAKE	ON	N	-12.0 to -12.5
5	REINSERT PENDULUM	OFF	A	-12.0 to -12.5
6	REINSERT PENDULUM	ON	A	-0.1 to +0.1
7	PITCH INHIBIT MOD	ON 1	HH	-21.4 to -22.5
8	PITCH INHIBIT MOD	OFF 1	HH	-0.9 to -1.8
9	PITCH INHIBIT MOD	OFF or ON 0	HH	-0.1 to +0.1
10	EXPLODER ARM	OFF	PP	+39.7 to +40.3
11	EXPLODER ARM	ON	PP	-39.7 to -40.3
12	NOT SNAKE	OFF	J3601-H	-0.1 to +0.1
13	NOT SNAKE	ON	J3601-H	-14.7 to -15.2
14	NOT 6°	OFF	J4201-E	-0.1 to +0.1
15	NOT 6°	ON	J4201-E	-14.7 to -15.2

(e) PANEL NO. 3 test point continuity. Connect ohmmeter between PANEL No. 3 test points and connector pins listed in Table XXIII. Set GYRO PENDULUM SIMULATOR SELECTOR to YAW RATE-ROLL PENDULUM and COURSE/DEPTH SIMULATOR SELECTOR to 0'. Maximum resistance shall be as specified.

4.5.3.14 Load resistors (search depth and ceiling mode)

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Table XX

PANEL NO. 2 Switch Continuity

Test	PANEL NO. 2 Switch	Switch Position	Connector Pin	Continuity Indication (Ohms)
1	C G ACTIVATE	OPEN	J3601-FF	500,000 minimum
2	C G ACTIVATE	GRD	J3601-FF	0.5 maximum
3	RECOVERY	ON	J3601-DD	0.5 maximum
4	RECOVERY	OFF	J3601-DD	500,000 minimum
5	R<B	OPEN	J4201-B	500,000 minimum
6	R<B	GRD	J4201-B	0.5 maximum
7	3 MINUTE SNAKE	OPEN	J3601-R	500,000 minimum
8	3 MINUTE SNAKE	GRD	J3601-R	0.5 maximum

Table XXI

Function Selector Switch Continuity

Test	FUNCTION SELECTOR Switch Position	Connector Pin	Maximum Resistance (Ohms)
1	DOPPLER LOGIC	J4201-L	0.5
2	SD NOT ACHIEVED - PASSIVE	J4201-D	0.5
3	ACTIVE	J4201-J	0.5
4	PITCH RANGE CONSTRAINT	J4201-A	0.5
5	BEAM WIDTH CONTROL	J4201-K	0.5
6	COMPUTER ENABLE	J3601-B	0.5
7	NOT AIR/ASROC SEARCH	J4201-H	0.5
8	AIR/ASROC SEARCH	J4201-E	0.5
9	-30 V LOG AMP	J4201-Z	0.5

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Table XXII

PANEL NO. 1 Test Point Continuity

Test	Panel No. 1 Test Points	Connector Pin	Maximum Resistance (Ohms)
1	INST COURSE GYRO	J3601-Y	0.5
2	INST ROLL RATE GYRO	J3601-W	0.5
3	INST YAW RATE GYRO	J3601-Z	0.5
4	INST PITCH RATE GYRO	J3601-AA	0.5
5	INST ROLL PENDULUM	J3601-X	0.5
6	INST PITCH PENDULUM	J3601-CC	0.5
7	+90 VDC OUT (F)	J4201-F	0.5
8	+90 VDC OUT (N)	J4201-N	0.5
9	CG 400~	P3602-E	0.5

Table XXIII

PANEL NO. 3 Test Point Continuity

Test	PANEL NO. 3 Test Points	Connector Pin	Maximum Resistance (Ohms)
1	YAW (Output)	J3601-XX	0.5
2	PITCH (Output)	J3601-WW	0.5
3	ROLL (Output)	J3601-VV	0.5
4	COURSE/DEPTH Pot	J4202-P	0.5
5	RATE GYRO	J4202-K	0.5
6	PENDULUM	J4202-L	0.5

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(a) Search depth load resistors. Connect resistance bridge between connector J4201 pins i (SEARCH DEPTH SENSOR) and g (FIRE CONTROL COMMON). Set TEST DRIVE switch to TEST. With SEARCH DEPTH switch set to positions listed in Table XXIV, measure resistance between pins. Resistance shall be as specified in Table XXIV.

Table XXIV

Search Depth Load Resistor Measurement

Test	SEARCH DEPTH Switch Position	Limits (Ohms)
1	1	170 to 172
2	2	284.5 to 287.5
3	3	340.2 to 343.8
4	7	507.3 to 512.7

(b) Ceiling mode load resistors. Connect resistance bridge between connector J4201 Pins n (CEILING MODE SENSOR) and g (FIRE CONTROL COMMON). With CEILING MODE switch set to positions listed in Table XXV, resistance indications shall be as specified in Table XXV.

Table XXV

Ceiling Mode Load Resistor Measurement

Test	CEILING MODE Switch Position	Limits (Ohms)
1	1	158.3 to 161.7
2	2	238.7 to 241.3
3	5	504.8 to 515.2

4.5.3.15 4.8 KC grounds isolation

(a) Connect ohmmeter to TB1, negative (-) lead to pin 15 (4.8 KC GRD) and positive (+) lead to pin 11 (PWR GRD) on Chassis No 3. Resistance indication shall be greater than 500,000 ohms.

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(b) Measure resistance at connector pins listed in Table XXVI. Resistance indications shall be as specified.

Table XXVI

4.8 KC Grounds Isolation

Test	Connector Pin	Connector Pin	Limits (Ohms)
1	J4201-P	J4202-J	100 to 150
2	J4201-P	P3602-B	500,000 min

4.5.3.16 400 Hz load resistors

(a) Using resistance bridge, measure resistance between connector J4202 pins A (400~ ∅ A) and B (400~ GROUND). Resistance indication shall be 61.5 to 68.5 ohms.

(b) Remove C. G. 400~ indicator lamp bulb from PANEL NO. 1. Using resistance bridge, measure load resistance between connector P3602 pins E (400~ ∅ A) and D (400~ GROUND). Resistance indication shall be 312 to 348 ohms. Replace C. G. 400~ indicator bulb.

4.5.3.17 Computer enable load resistor

(a) Remove COMPUTER ENABLE indicator lamp bulb. Set FUNCTION SELECTOR to -30 V LOG AMP. Using resistance bridge, measure resistance between J4201-P (GRD) and J3601-B (COMPUTER ENABLE). Resistance indication shall be 58.5 to 65.5 ohms.

(b) Replace COMPUTER ENABLE indicator lamp bulb.

4.5.4 Visual inspections

4.5.4.1 Identification marking

(a) The nomenclature, control callout and identification data shall be correct, clear, and legible. (Reference: DL 2540360.)

4.5.4.2 Interchangeability

(a) The connector receptacles shall be correct and capable of accepting the mating units without resort to selection, modification, or fitting. (Reference: DL 2540360.)

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4.5.5 Process verification

4.5.5.1 Criteria stipulated in this paragraph are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection by the contractor.

4.5.5.2 Lacing

(a) Lacing and cable configuration shall be in accordance with precedures approved by the procuring activity. Specific method of verifying the adequacy of cable lacing shall include visual and dimensional verification.

4.5.5.3 Soldering

(a) Soldered joints shall be in accordance with good commercial practices. Unsoldered connections or those of a "cold" solder or "rosin" joint shall be considered defective.

4.5.5.4 Cleanliness

(a) Presence of foreign material in assembly shall classify unit as defective. Foreign material includes anything not intended to be present.

4.5.6 Packaging and packing inspections. Packaging, packing, and marking shall comply with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level C.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity and may conform to the supplier's commercial practice.

5.2 Packing. Packing shall be Level C as specified.

5.2.1 Level A. Not applicable.

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5.2.2 Level B. Not applicable.

5.2.3 Level C. The test set, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall conform with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use and description

6.1.1 Intended use. The test set is intended for use in a manufacturing facility for quality conformance testing of automatic pilot for Torpedo Mk 46.

6.1.2 Description

6.1.2.1 Mechanical. The test set is a rectangular cabinet containing five test panels and three rack-mounted power supplies.

6.1.2.2 Electrical. The test set simulates the electrical inputs and outputs of the Automatic Pilot.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required data.

(c) Any exceptions to this specification.

(d) All additional samples required shall be provided and inspected at the expense of the supplier when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions. For the purpose of this specification the following definitions have been established:

6.3.1 Operating. The condition of the test set when it is performing its required function.

6.3.2 Temperature. The temperature of the environment to which the test set may be exposed, expressed in degrees Fahrenheit (°F).

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6.3.3 Pressure. The pressure of the environment to which the test set may be exposed, expressed in pounds per square inch absolute (psia).

6.3.4 Humidity. The relative humidity of the environment to which the test set is exposed, expressed as a percent.

Preparing Activity
Navy - OS
Project No. 1355-N292

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

AUTOMATIC PILOT TEST SET

DL 2540360

(For Torpedo Mk 46)

EQUIPMENT STATUS			
CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

AUTOMATIC PILOT TEST SET

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.5 of this specification. When completed with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the test set bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.5 shall be completed and the results recorded or verified in this report form prior to acceptance of the test set. Unless otherwise noted the sequence of testing, recording, and verification, as performed in 4.5, is mandatory and must be performed in the order shown.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading will be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column, indicating that the requirement has been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Test equipment	4.5.2	In calibration	_____
30.2 Test procedure	4.5.3		
30.2.1 Power supply tests	4.5.3.1		
+40 VDC	4.5.3.1.2		
		(a) +39.9 to +40.1 vdc	_____
		(b) 480 to 520 ma	_____

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30.2.1 (cont'd)	Reference	Limits	Reading
-40 VDC	4.5.3.1.3		
	(a)	-39.9 to -40.1	_____
-15 VDC	4.5.3.1.4		
	(a)	-14.9 to -15.1 vdc	_____
	(b)	238 to 262 ma	_____
+15 VDC	4.5.3.1.5		
Adjustment Range	(a)	+14.25 vdc max to +15.75 vdc min	_____ _____
	(b)	+14.9 to +15.1 vdc	_____
	(c)	71 to 77 ma	_____
+30 VDC	4.5.3.1.6		
Adjustment Range	(a)	-28.5 vdc max to -31.5 vdc min	_____ _____
	(b)	-29.9 to -30.1 vdc	_____
	(c)	142 to 157 ma	_____
+125 VDC	4.5.3.1.7		
	(b)	+123 to +127 vdc	_____
+90 VDC IN(S)	(d)	+107 to +113 vdc	_____
+90 VDC IN(T)	(d)	+107 to +113 vdc	_____
30.2.2 Internal test set voltages	4.5.3.2		
	(a)	+4.6 to +5.4 vdc	_____
	(b)	-18.7 to -21.3 vdc	_____

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	Reference		Limits	Reading
30.2.3 Test point wiring	4.5.3.3			
+40 VDC TS		(a)	+39.5 to +40.5 vdc	_____
+15 VDC TS			+14.0 to +16.0 vdc	_____
-15 VDC TS			-14.5 to -15.5 vdc	_____
-30 VDC TS			-29.0 to -31.0 vdc	_____
-40 VDC TS			-39.5 to -40.5 vdc	_____
+150 VDC TS		(b)	+140 to +160 vdc	_____
30.2.4 Indicator lamp pretest	4.5.3.4			
		(b)	All indicators illuminate	_____
30.2.5 4.8 KC power supply	4.5.3.5			
Amplitude adjustment	4.5.3.5.1			
		(e)	24.0 to 28.0 v rms	_____
4.9 KC adjustment	4.5.3.5.2			
		(b)	4890 Hz to min	_____
			4910 Hz max	_____
4.8 KC adjustment	4.5.3.5.3			
		(a)	4790 Hz to min	_____
			4810 Hz max	_____
4.7 KC adjustment	4.5.3.5.4			
		(a)	4690 Hz to min	_____
			4710 Hz max	_____
4.8 KC readjustment	4.5.3.5.5			
		(a)	4800 ±2 Hz	_____

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		Reference	Limits	Reading
30.2.6	Gyro pendulum assembly simulator	4.5.3.6		
		(c)	25.0 mv rms max	_____
		(e)	11.55 to 11.65 v rms	_____
		(f)	11.55 to 11.65 v rms	_____
		(h)	25.0 mv rms max	_____
		(i)	11.30 to 11.40 v rms	_____
		(j)	11.30 to 11.40 v rms	_____
		(k)	0.5 percent max	_____
	Distortion	(l)		_____
	(k) - (l)		0.5 percent max	_____
	Test	(m)		
	1		-2° to +2°	_____
	2		178° to 182°	_____
	3		-2° to +2°	_____
	4		178° to 182°	_____
	5		178° to 182°	_____
	6		-2° to +2°	_____
	7		178° to 182°	_____
	8		-2° to +2°	_____
	9		-2° to +2°	_____
	10		178° to 182°	_____
	11		-2° to +2°	_____
	12		178° to 182°	_____

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		Reference	Limits	Reading
30.2.6 (Cont'd)				
		(n)	in phase	
		(o)	180° out of phase	
	Channel A	(p)	approx. zero volts	
	Channel B		approx. zero volts	
30.2.7	Course depth simulator	4.5.3.7		
		(e)	25 mv rms max	
		(f)	25.9 to 26.1 v rms	
		(i)	less than 6 ohms	
		(j)	7425 to 7575 ohms	
		(k)	95 to 105 ohms	
	Test	(m)		
	1		-29.0 to -30.0 vdc	
	2		-1.0 to +1.0 vdc	
	3		-1.0 to +1.0 vdc	
	4		-29.0 to -30.0 vdc	
	5		-29.0 to -30.0 vdc	
	6		-29.0 to -30.0 vdc	
	Test	(n)		
	1		-29.0 to -30.0 vdc	
	2		-1.0 to +1.0 vdc	
	3		-1.0 to +1.0 vdc	
	4		-1.0 to +1.0 vdc	

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		Reference	Limits	Reading
30.2.7 (Cont'd)				
	5		-29.0 to -30.0 vdc	_____
	6		-29.0 to -30.0 vdc	_____
	Test	(o)		
	1		-29.0 to-30.0 vdc	_____
	2		-1.0 to +1.0 vdc	_____
	3		-1.0 to +1.0 vdc	_____
	4		-1.0 to +1.0 vdc	_____
	5		-1.0 to +1.0 vdc	_____
	6		-29.0 to -30.0 vdc	_____
30.2.8 Fire control		4.5.3.8		
Adjustment Range		(b)	+18 vdc max	_____
			+30 vdc min	_____
	Test	(f)		
	1		+2.80 to +2.90 vdc	_____
	2		+5.60 to +5.80 vdc	_____
	3		+8.40 to +8.70 vdc	_____
	4		+11.2 to +11.6 vdc	_____
	5		+14.0 to +14.5 vdc	_____
	6		+16.9 to +17.3 vdc	_____
	7		+19.8 to +20.2 vdc	_____
		(g)	-1.0 to +1.0 vdc	_____
		(h)	+11.2 to +11.6 vdc	_____
		(i)	+19.8 to +20.2 vdc	_____
		(k)	+19.8 to +20.2 vdc	_____

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30.2.8 (Cont'd)	Reference	Limits	Reading
SEARCH DEPTH indicator		illuminates	
Voltage drops to zero		7 seconds max	
Test	(n)		
1		+3.90 to +4.10 vdc	
2		+7.85 to +8.10 vdc	
3		+11.8 to +12.2 vdc	
4		+15.8 to +16.2 vdc	
5		+19.8 to +20.2 vdc	
	(o)	+19.8 to +20.2 vdc	
	(p)	-1.0 to +1.0 vdc	
	(r)	+19.8 to +20.2 vdc	
CEILING MODE indicator		illuminates	
Voltage drops to zero		7 seconds max	
	(t)	+19.8 to +20.2 vdc	
	(u)	1900 to 2100 ohms	
30.2.9 Functional check	4.5.3.9		
MOD 2 indicator	(a)	extinguished	
Test	(b)		
1 SNAKE SEARCH		illuminates	
2 POWER REDUCTION		illuminates	
3 DOPPLER LOGIC		illuminates	
4 SD NOT ACHIEVED - PASSIVE		illuminates	
5 ACTIVE		illuminates	

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30.2.9 Functional check (Cont'd)	Reference	Limits	Reading
6 PITCH RANGE CONSTR		illuminates	
7 BEAM WIDTH CONTROL		illuminates	
8 COMPUTER ENABLE		illuminates	
9 NOT AIR/ASROC SEARCH		illuminates	
10 AIR/ASROC SEARCH		illuminates	
11 ODN TRIGGER		illuminates	
CONTINUITY	(c)	illuminates	
30.2.10 Autopilot outputs	4.5.3.10		
Test	(a)		
1 CW		illuminated	
CCW		extinguished	
2 CW		extinguished	
CCW		illuminated	
3 DOWN		illuminated	
UP		extinguished	
4 DOWN		extinguished	
UP		illuminated	
5 RIGHT		illuminated	
LEFT		extinguished	
6 RIGHT		extinguished	
LEFT		illuminated	
	(b)		
YAW (output) MOD 1		900 to 920 ohms	
MOD 0		332 to 368 ohms	
PITCH (output) MOD 1		900 to 920 ohms	
MOD 0		332 to 368 ohms	
ROLL (output) MOD 1		900 to 920 ohms	
MOD 0		332 to 368 ohms	

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		Reference	Limits	Reading
30.2.11	Pitch and yaw commands	4.5.3.11		

Pitch commands voltages:

PITCH COMMANDS

0	MOD 1	(b)	-24 to -26 mv dc	_____
	MOD 0		-5 to +5 mv dc	_____
1 UP		(c)	+1.23 vdc max	_____
			+2.22 vdc min	_____
2 UP		(e)	+2.56 vdc max	_____
			+3.26 vdc min	_____
1 DOWN		(g)	-1.23 vdc max	_____
			-2.22 vdc min	_____
2 DOWN		(i)	-2.56 vdc max	_____
			-3.26 vdc min	_____

Yaw commands voltages:

YAW COMMANDS

0	MOD 1	(k)	-24 to -26 mv dc	_____
	MOD 0		-5 to +5 mv dc	_____
1 LEFT		(l)	-1.23 vdc max	_____
			-2.22 vdc min	_____
2 LEFT		(n)	-2.56 vdc max	_____
			-3.26 vdc min	_____
1 RIGHT		(p)	+1.23 vdc max	_____
			+2.22 vdc min	_____
2 RIGHT		(r)	+2.56 vdc max	_____
			+3.26 vdc min	_____
MOD 2 indicator		(u)	illuminated	_____
MOD 2 indicator			illuminated	_____
YAW COMMANDS 2 RIGHT		(v)	+1.38 to +1.44 vdc	_____
PITCH COMMANDS 2 UP		(w)	+1.38 to +1.44 vdc	_____

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30.2.11 (Cont'd)	Reference	Limits	Reading
Resistance	(x)		
(LEFT to RIGHT)		25.3K to 25.8K	_____
(DOWN to UP)		25.3K to 25.8K	_____
PASSIVE indicator	(y)	illuminates	_____
SD NOT ACHIEVED - PASSIVE	(z)	-14.7 to -15.2 vac	_____
UP (MOD 1)	(aa)	-5 to +5 mv dc	_____
30.2.12 Voltage wiring check	4.5.3.12		
Test	(b)		
1		-0.1 to +0.1 vdc	_____
2		-12.0 to -12.5 vdc	_____
3		-0.1 to +0.1 vdc	_____
4		-12.0 to -12.5 vdc	_____
5		-12.0 to -12.5 vdc	_____
6		-0.1 to +0.1 vdc	_____
7		-21.4 to -22.5 vdc	_____
8		-0.9 to -1.8 vdc	_____
9		-0.1 to +0.1 vdc	_____
10		+39.7 to +40.3 vdc	_____
11		-39.7 to -40.3 vdc	_____
12		-0.1 to +0.1 vdc	_____
13		-14.7 to -15.2 vdc	_____
14		-0.1 to +0.1 vdc	_____
15		-14.7 to -15.2 vdc	_____
Resistance	(c)		
NOT 10° switch OFF		0.5 ohm max	_____
NOT 10° switch ON		500K min	_____

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	Reference	Limits	Reading
30.2.13 Continuity tests	4.5.3.13		
Test	(a)		
1		500,000 ohms min	_____
2		0.5 ohm max	_____
3		0.5 ohm max	_____
4		500,000 ohms min	_____
5		500,000 ohms min	_____
6		0.5 ohm max	_____
7		500,000 ohms min	_____
8		0.5 ohm max	
Test	(b)		
1		0.5 ohm max	_____
2		0.5 ohm max	_____
3		0.5 ohm max	_____
4		0.5 ohm max	_____
5		0.5 ohm max	_____
6		0.5 ohm max	_____
7		0.5 ohm max	_____
8		0.5 ohm max	_____
9		0.5 ohm max	_____
	(c)	900 to 1100 ohms	_____
Test	(d)		
1		0.5 ohm max	_____
2		0.5 ohm max	_____

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		Reference	Limits	Reading
30.2.13	(Cont'd)			
	3		0.5 ohm max	
	4		0.5 ohm max	
	5		0.5 ohm max	
	6		0.5 ohm max	
	7		0.5 ohm max	
	8		0.5 ohm max	
	9		0.5 ohm max	
	Test	(e)		
	1		0.5 ohm max	
	2		0.5 ohm max	
	3		0.5 ohm max	
	4		0.5 ohm max	
	5		0.5 ohm max	
	6		0.5 ohm max	
30.2.14	Load resistors (search, depth, and ceiling mode)	4.5.3.14		
	Test	(a)		
	1		170 to 172 ohms	
	2		284.5 to 287.5 ohms	
	3		340.2 to 343.8 ohms	
	4		507.3 to 512.7 ohms	
	Test	(b)		
	1		158.3 to 161.7 ohms	
	2		238.7 to 241.3 ohms	
	3		504.8 to 515.2 ohms	

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		Reference	Limits	Reading
30.2.15	4.8 KC grounds isolation.	4.5.3.15		
		(a)	500,000 ohms min	_____
Test		(b)		
1			100 to 150 ohms	_____
2			500,000 ohms min	_____
30.2.16	400 Hz load resistors	4.5.3.16		
		(a)	61.5 to 68.5 ohms	_____
		(b)	312 to 348 ohms	_____
30.2.17	Computer enable load resistor	4.5.3.17		
		(a)	58.5 to 65.5 ohms	_____
30.3	Visual inspections	4.5.4		
30.3.1	Indentification marking	4.5.4.1		
	Nomenclature, control and identification callouts	4.5.4.1(a)		_____
30.3.2	Interchangeability	4.5.4.2		
	Connector receptacles	4.5.4.2(a)		_____
30.4	Process verification	4.5.5		
30.4.1	Lacing	4.5.5.2		
	Lacing and cable configuration	4.5.5.2(a)		
30.4.2	Soldering	4.5.5.3		
	Soldered joints	4.5.5.3(a)		_____

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	Reference	Limits	Reading
30.4.3 Cleanliness	4.5.5.4		
Foreign material	4.5.5.4(a)		_____
30.5 Packaging and packing inspections	4.5.6		

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-A-82450B(OS) Automatic Pilot Test Set

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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Commanding Officer
Naval Ordnance Station
Standardization Division (QA3)
Indian Head, MD 20640

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